

MuSR Blackboard Adaptive Communication ($q = 3$) DeepInfra Analysis Report

OpenAI analysis handoff

2026-09-07

Executive summary

This report summarizes the first analysis of the archive `musr_blackboard_adaptive_communication_q3_deepinfra`. The central result is that the adaptive controller is *much weaker as an adversarial steering mechanism* than the earlier truthful-report-only controller, even though it intervenes more often. The most plausible explanation is that the new controller spends a large fraction of active rounds asking questions or issuing directives rather than injecting strategically selected truthful evidence. This appears to trade raw steering power for information-seeking and coordination.

The archive is usable but incomplete: 723/750 planned episodes are present, with 27 missing episode keys and no failed/aborted episodes. This is sufficient for a strong diagnostic read, but final paper numbers should eventually be refreshed on the completed archive.

1 Study status and configuration

1.1 Validation summary

According to the analysis validation file, the study contains:

- 3/3 expected run configurations,
- 75/75 scientific cells,
- 723/750 completed episodes,
- 7,230 round records and 173,520 micro-slot records,
- no duplicate identities and no failed or aborted episodes.

The reported validation status is **INVALID** only because 27 target episode keys are missing while aggregation was allowed to continue under `--allow-incomplete`.

1.2 Main experimental parameters

The resolved run configurations indicate the following shared design for the three arms (no control, truth control, false control):

Parameter	Value
Task family	MuSR team allocation
Task id	<code>task_001</code>
Population size	$N = 24$
Population rounds / horizon	10
Social mode	public blackboard
Social group size	$q = 3$ visible messages
Sensor sample size	$q_c = 12$
Controller policy	soft-target, $\beta = 4.0$, threshold $\theta = 0.5$
Persistence grid	$\rho \in \{0.70, 0.775, 0.85, 0.925, 1.00\}$
Budget grid	$b \in \{3, 6, 9, 12, 15, 18, 21\}$
Board lifetime	1 round
Participant requests	enabled
Controller mode	<code>adaptive_communication</code>
Controller timing	dawn only
Allowed controller modes	REQUEST, REPORT, DIRECTIVE
Prompt family / version	<code>relational_blackboard_ballot@4</code>
LLM provider	DeepInfra / <code>deepseek-ai/DeepSeek-V4-Flash</code>
Repetitions	10 per cell

2 Headline behavioral findings

The key outcome-level comparison against the previous truthful-report-only study is shown in Table 1. The adaptive controller is clearly weaker, especially for false control.

Table 1: Headline aggregate comparison with the previous report-only study. Values are study-level averages quoted from the corresponding analyses.

Arm / controller	p_{truth} final	p_{false} final	False plurality rate	Mean χ
False control, report-only	0.445	0.397	42.3%	0.088
False control, adaptive	0.598	0.246	16.3%	0.031
Truth control, report-only	—	—	—	0.027
Truth control, adaptive	—	—	—	0.013

The most important qualitative conclusions are:

1. **Adaptive false control no longer captures the population reliably.** The false target ends as the final plurality in only about 16.3% of episodes, versus about 42.3% for the earlier report-only controller.
2. **The adaptive controller acts more often but steers less.** For false control, $P(U = 1) \approx 0.668$ here versus roughly 0.594 previously. Nevertheless the mean target-share response per active round drops from about +0.053 to +0.010.
3. **The realization of control matters.** Keeping the intervention variable binary is scientifically sensible: the main change is not whether the controller acts, but how it realizes an active intervention.

3 Adaptive mode selection

A first distinctive feature of this archive is that the controller meaningfully uses all three communication acts on active rounds. Figure 1 shows the average active-round mode shares.

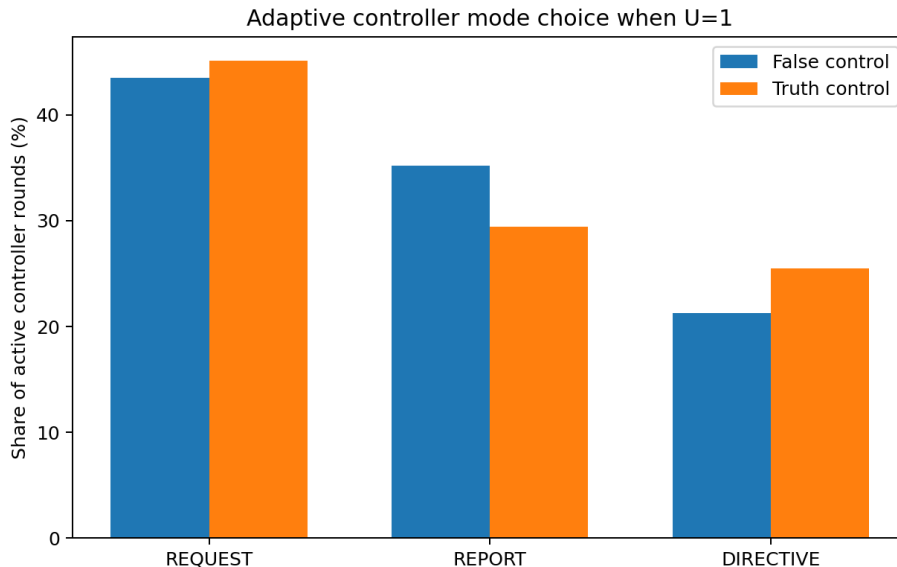


Figure 1: Adaptive controller mode choice conditional on an active intervention $U = 1$.

On active false-control rounds the controller chooses approximately 43.5% REQUEST, 35.2% REPORT, and 21.3% DIRECTIVE. For truth control the proportions are similar: 45.1% REQUEST, 29.4% REPORT, and 25.5% DIRECTIVE. Thus the controller is not simply reverting to the old report-only behavior.

4 Immediate effect of each communication mode

The mode-conditioned same-round target-share changes are especially informative. Figure 2 summarizes the estimated immediate effect of each adaptive communication act.

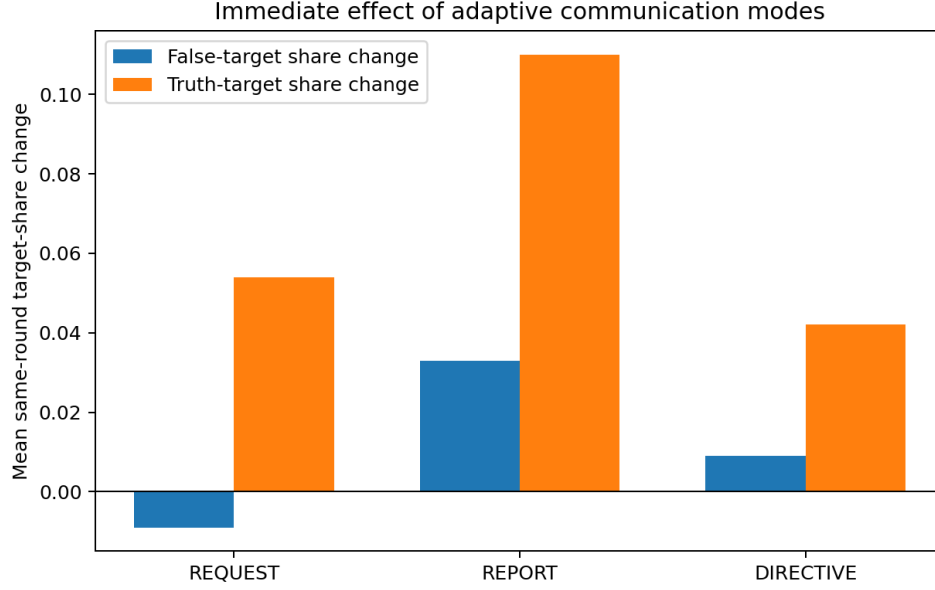


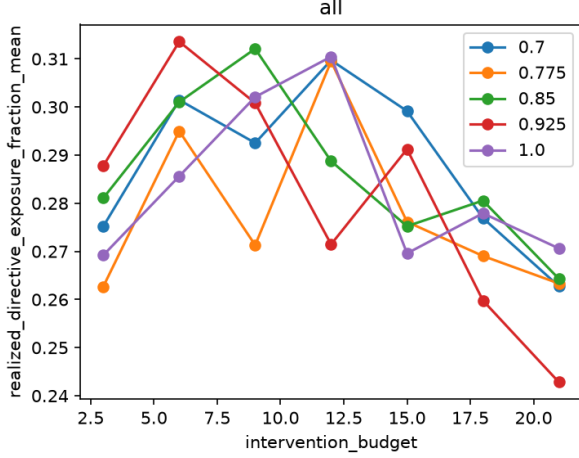
Figure 2: Mean same-round target-share change by adaptive communication mode. For false control the target is the configured false allocation; for truth control it is the true allocation.

For false control, REPORT is the only clearly strong steering act, with mean same-round false-target change around +0.033. DIRECTIVE is weaker at about +0.009, while REQUEST is actually counterproductive at about -0.009 . In fact, under a false-target REQUEST the truth share rises by roughly +0.026 on average. This is the clearest signal that information-seeking behavior can undermine adversarial selective disclosure.

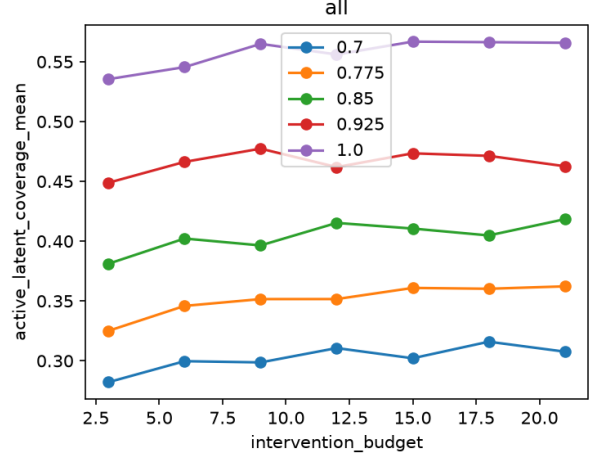
For truth control, all three acts help the truth, but again REPORT is strongest: approximately +0.110 for REPORT, +0.054 for REQUEST, and +0.042 for DIRECTIVE.

5 Controller intensity and realized budget

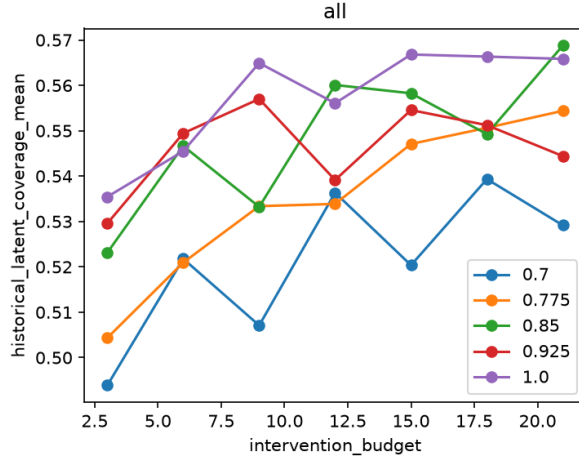
A second major difference is that nominal intervention budget b is no longer equal to the realized number of controller posts, because REQUEST and DIRECTIVE typically use only one message while REPORT can consume the larger information budget. Figure 3 shows the supplied budget-level diagnostics from the archive.



(a) Realized directive exposure fraction



(b) Active evidence coverage



(c) Historical evidence coverage

Figure 3: Archive-supplied budget diagnostics for the adaptive-communication study.

The analysis indicates that false control now averages only about 3.13 controller posts per active round, and truth control about 3.71, far below the roughly 12 posts per active round in the earlier report-only study. For false control, the realized posts per active round rise only approximately as

$$1.66, 2.70, 3.24, 3.48, 3.45, 3.73, 3.73$$

across nominal budgets $b = 3, 6, 9, 12, 15, 18, 21$. Thus the actuator effectively saturates around $b \approx 12$.

Figure 4 summarizes the main comparison with the earlier report-only controller for false control.

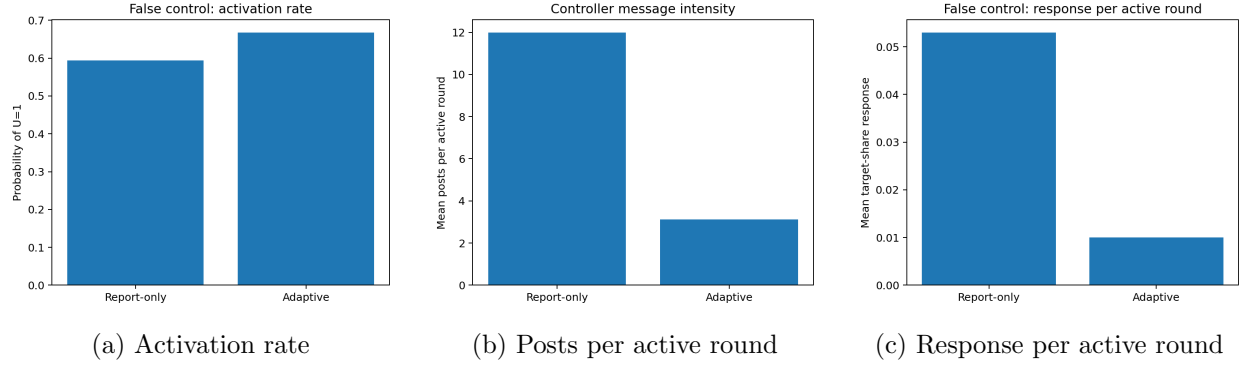


Figure 4: False-control comparison between the earlier report-only controller and the present adaptive controller.

This makes the main control-efficiency message very clear: *more interventions do not automatically mean more effective control*. The controller acts more often but spends a substantial portion of active rounds using weaker or even corrective communication modes.

6 Information and susceptibility metrics

6.1 Susceptibility

The susceptibility signal remains one of the most informative state-local summaries. Figure 5 compares the mean χ values against the earlier report-only study.

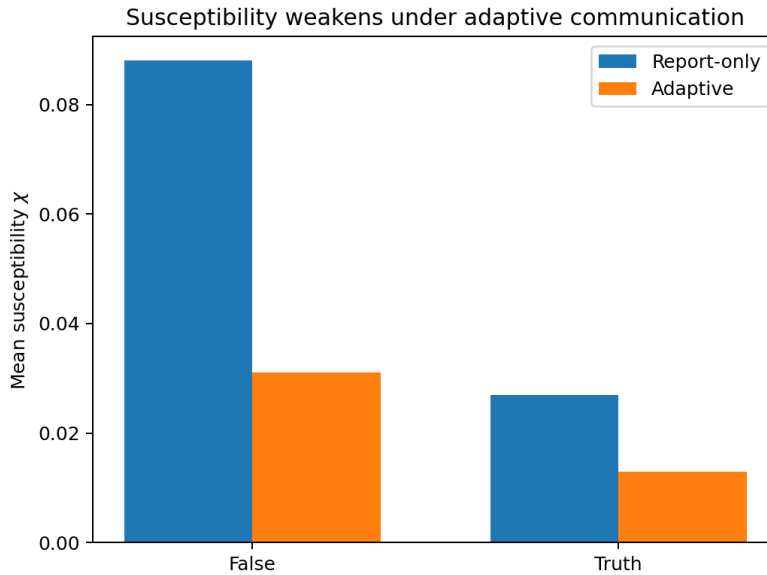


Figure 5: Mean susceptibility comparison between report-only and adaptive communication controllers.

For adaptive false control, the mean susceptibility drops to about 0.031 from roughly 0.088 in the report-only study. Truth control also weakens, from approximately 0.027 to 0.013. In cellwise

terms, only 10/35 adaptive false-control cells have a bootstrap confidence interval for χ entirely above zero, compared with 26/35 previously.

6.2 Transfer information

The raw conditional transfer information remains large, but the same finite-sample null issue observed in the previous study is still present. Figure 6 shows the aggregate comparison.

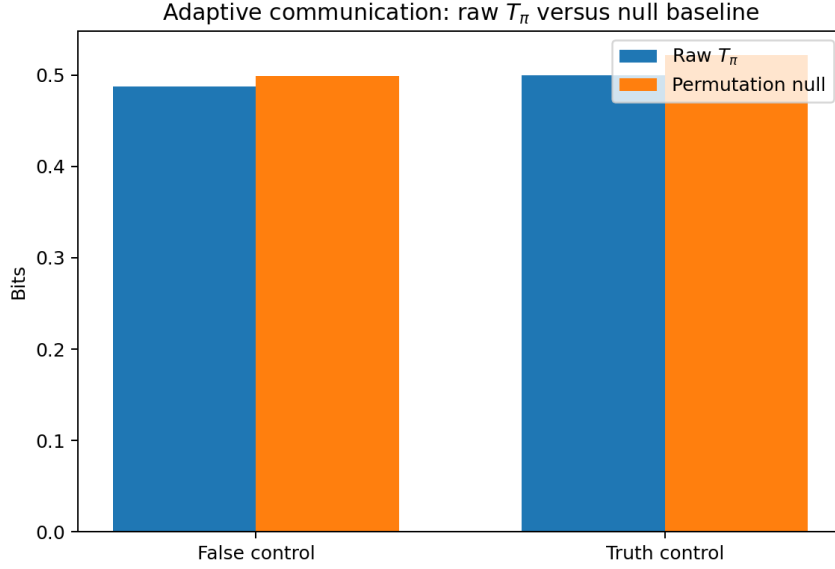


Figure 6: Raw T_π against the permutation null baseline.

Aggregate values are approximately:

$$T_\pi^{\text{false}} \approx 0.487, \quad T_{\text{null}}^{\text{false}} \approx 0.499,$$

$$T_\pi^{\text{truth}} \approx 0.500, \quad T_{\text{null}}^{\text{truth}} \approx 0.522.$$

Thus the null-corrected excess is slightly negative in both arms, and there are no cells with permutation $p < 0.05$. As in the previous study, raw T_π should therefore not be interpreted without its null reference. This also means that downstream quantities such as η_{IF} should be treated cautiously when they inherit the same finite-sample baseline inflation.

6.3 Thermodynamic efficiency

The thermodynamic efficiency η_{th} remains unsupported here for the same reason as in the earlier blackboard analyses: the actuator-specific calibration needed for hJ_c has not yet been supplied for this communication mechanism. The absence of η_{th} values should be kept explicit rather than filled heuristically.

7 State-resolved phase maps: current archive issue

The archive includes the usual $x \times b$ plot files for T_π , χ , η_{IF} , η_{IR} , and occupancy. However, all of these rendered images are gray placeholders labeled `structural cell not run`. Example files

are included in Appendix A. This appears to be the same packaging/export problem seen in the previous study: the archive contains state-resolved estimate tables, but the bundled phase-map images were exported from records tagged as unsupported.

For the present report, the safest course is therefore:

- use the aggregate and communication-mechanism conclusions as primary evidence;
- treat the bundled gray phase-map figures as an archive/export issue rather than a scientific result;
- regenerate the state-local surfaces from the estimator tables in a fully parquet-enabled environment if the paper draft needs the full $x \times b$ surfaces.

8 Interpretation

The central scientific interpretation is:

adaptive communication trades raw steering power for information-seeking and coordination.

This trade-off is strongly asymmetric. For truth control, the system still behaves sensibly: REPORT, REQUEST, and DIRECTIVE all move the population in the intended direction, though REPORT is best. For false control, by contrast, REQUEST often acts as a corrective mechanism because it encourages the population to retrieve missing evidence that undermines the selectively disclosed false narrative.

This result is quite valuable for the broader project. It suggests that the *communication protocol itself* can protect a multi-agent population against adversarial steering, even when the controller remains binary and obeys the same sensing/actuation formalism. In other words, the efficiency of control is not only a function of how often the controller acts, but also of how much the population can use questions and peer evidence to repair partial information.

9 Recommended next steps

1. Complete the 27 missing adaptive episodes before freezing paper-level numbers.
2. Run the matched factorial comparison we have been designing: participant questions ON/OFF and controller directives ON/OFF, while keeping the same binary controller variable $U \in \{0, 1\}$.
3. Regenerate the $x \times b$ phase surfaces from the state-local estimator tables in a parquet-capable environment so that the final paper shows non-placeholder phase maps.
4. Compare adaptive communication directly against truthful report-only control under strictly matched prompt/version and initialization settings.

A Archive-supplied phase-map figures

The following figures are included for completeness. They illustrate the current export problem: the files exist, but all cells are rendered as unsupported placeholders.

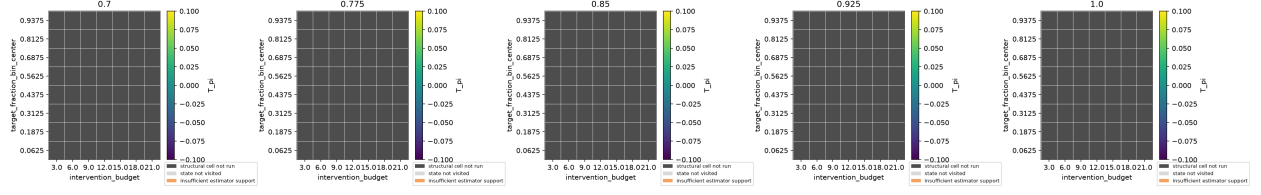


Figure 7: Bundled $T_\pi(x, b)$ phase maps from the archive.

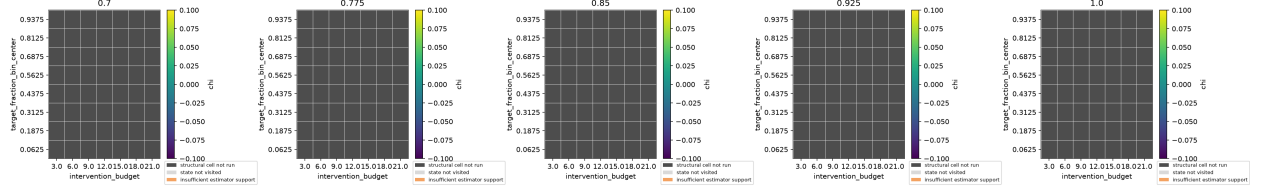


Figure 8: Bundled $\chi(x, b)$ phase maps from the archive.

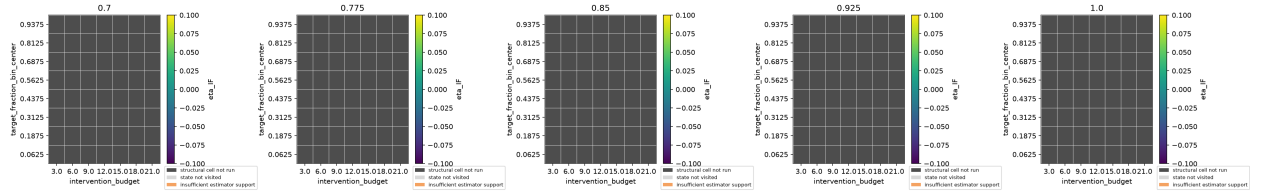


Figure 9: Bundled $\eta_F(x, b)$ phase maps from the archive.

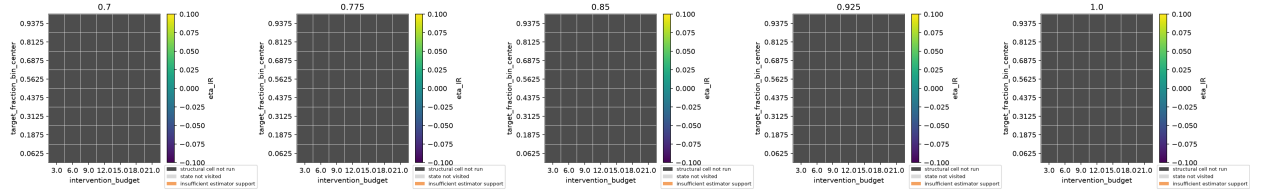


Figure 10: Bundled $\eta_R(x, b)$ phase maps from the archive.